

Optimization 1

ISE 5406

Lecture 2

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Reading for this lecture

- Chapter 2 of Bazaraa, Sherali, and Shetty (2006)
- Parts of Chapters 2 and 3 of Boyd and Vandenberghe

Outline

Linear Algebra Review

Convex Sets

Examples of Convex Sets

Operations Preserving Convexity of Sets

Convex and Concave Functions

Examples of Convex and Concave Functions

Convex Hulls

Carathéodory's Theorem

Introduction to Convex Analysis

Roadmap

Linear Algebra Review

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Definitions: Linear Algebra

- Euclidean Space ($\mathbb{R}^n$): The collection of all vectors of dimension n .

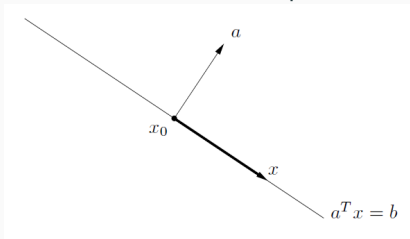
Definitions: Linear Algebra

- Euclidean Space ($\mathbb{R}^n$): The collection of all vectors of dimension n .
- Hyperplane: A set of the form

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = b\},$$

where $\mathbf{a} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $b \in \mathbb{R}$.

- A line when $n = 2$ and a plane when $n = 3$.



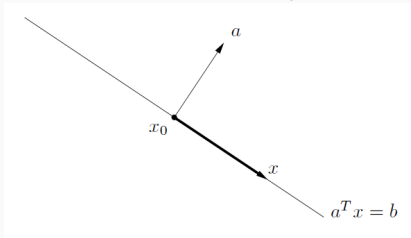
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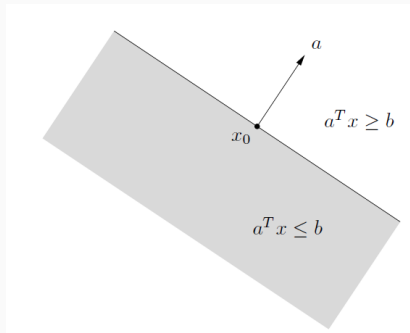
- Set of points $\mathbf{x}$ with constant inner product (b) to given vector $\mathbf{a}$.
- A hyperplane divides $\mathbb{R}^n$ into two **halfspaces**.

Linear Algebra: Definitions

- Halfspaces:

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \leq b\} \text{ and } \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \geq b\},$$

where $\mathbf{a} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $b \in \mathbb{R}$.



Linear Algebra: Definitions

Consider vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ in $\mathbb{R}^n$.

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- **Linear Combination:** A vector $\mathbf{b} \in \mathbb{R}^n$ is said to be a *linear* combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$, if

$$\mathbf{b} = \sum_{j=1}^k \lambda_j \mathbf{a}_j$$

where $\lambda_j \in \mathbb{R}$ for $j = 1, \dots, k$.

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where $\lambda_j \geq 0$ for $j = 1, \dots, k$.

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Linear Algebra: Definitions

- Linear/Affine Subspace ($S_L \subseteq \mathbb{R}^n$): If $\mathbf{a}_1$ and $\mathbf{a}_2 \in S_L$, then every linear/affine combination of $\mathbf{a}_1$ and $\mathbf{a}_2 \in S_L$.

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- **Linear Independence:** Vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ in $\mathbb{R}^n$ are said to be linearly independent if

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- **Span**: Vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ in $\mathbb{R}^n$ are said to **span** $\mathbb{R}^n$ if *any* vector in $\mathbb{R}^n$ can be represented as their linear combination.

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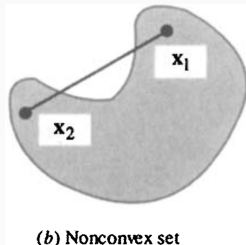
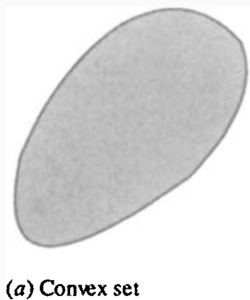
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 6. $\{\mathbf{x} \in \mathbb{R}_+^2 : x_1 x_2 = 0\}$
 7. $\{\mathbf{x} \in \mathbb{R}^2 : x_1^2 - x_1 x_2^2 + x_2^4 + 1 \geq 0\}$

Examples of Convex Sets: Polyhedra and Polyhedral Cones

Polyhedral Set or Polyhedron: Intersection of a finite number of halfspaces; $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq \mathbf{b}\}$, where $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$.

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Polyhedral Cone: Intersection of a finite number of halfspaces, whose hyperplanes pass through the origin, i.e., $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq \mathbf{0}\}$.

- Special case of polyhedral sets.

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Convex Cone: A convex set X with the additional property

$$\mathbf{x} \in X \text{ and } \lambda \geq 0 \implies \lambda\mathbf{x} \in X.$$

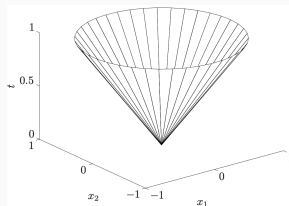
Polyhedral cones are a special case of convex cones.

Examples of Convex Sets: Ellipsoids and Convex Cones

Examples of convex cones:

- Lorentz or Second-Order Cone:

$$\mathcal{L}^{n+1} := \{(\mathbf{x}, t) \in \mathbb{R}^n \times \mathbb{R} : \|\mathbf{x}\|_2 \leq t\}.$$



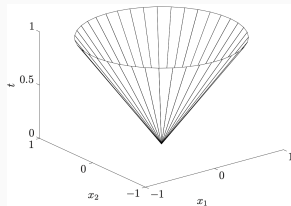
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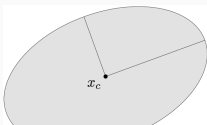
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Other examples of convex sets:

- **Euclidean Ball:** $\mathcal{B}(\mathbf{x}_c, r) := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_c\|_2 \leq r\}$.
- **Ellipsoid:** $\mathcal{E} := \{\mathbf{x}_c + A\mathbf{u} : \|\mathbf{u}\|_2 \leq 1\} \subseteq \mathbb{R}^n$, where $A \in \mathcal{S}_+^n$.



Operations Preserving Convexity of Sets (Not Exhaustive)

Lemma

Let S, S_1, S_2 be convex sets in $\mathbb{R}^n$. Then, the following sets are convex:

1. Intersection: $S_1 \cap S_2$.

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Let S, S_1, S_2 be convex sets in $\mathbb{R}^n$. Then, the following sets are convex:

1. Intersection: $S_1 \cap S_2$.
2. Minkowski Addition: $S_1 \oplus S_2 := \{\mathbf{x}_1 + \mathbf{x}_2 : \mathbf{x}_1 \in S_1, \mathbf{x}_2 \in S_2\}$.
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6. Cartesian Product: $S_1 \times S_2 := \{(\mathbf{x}_1, \mathbf{x}_2) : \mathbf{x}_1 \in S_1, \mathbf{x}_2 \in S_2\}$.
7. Coordinate Projection: $\text{proj}_{\mathbf{x}}(S) := \{\mathbf{x} : (\mathbf{x}, \mathbf{y}) \in S \text{ for some } \mathbf{y}\}$.

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7. Coordinate Projection: $\text{proj}_{\mathbf{x}}(S) := \{\mathbf{x} : (\mathbf{x}, \mathbf{y}) \in S \text{ for some } \mathbf{y}\}$.
8. Affine Image: $\{A\mathbf{x} + \mathbf{b} : \mathbf{x} \in S\}$ for any $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$.
9. Affine Pre-image: $\{\mathbf{x} : A\mathbf{x} + \mathbf{b} \in S\}$ for some $A \in \mathbb{R}^{n \times m}$, $\mathbf{b} \in \mathbb{R}^n$.

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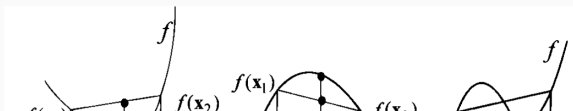
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Examples of Convex and Concave Functions

Examples of convex functions:

- Exponential function: $\exp(x)$ on $X = \mathbb{R}$.
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Level Sets of Convex Functions are Convex Sets

(Lower) Level Set: Let S be a convex set and $f : S \rightarrow \mathbb{R}$ be a convex function. Then, for any $\alpha \in \mathbb{R}$, the α -level set of f is defined as

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Convex Hulls

Carathéodory's Theorem

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for **some** positive integer k and points $\mathbf{x}_1, \dots, \mathbf{x}_k \in S$.

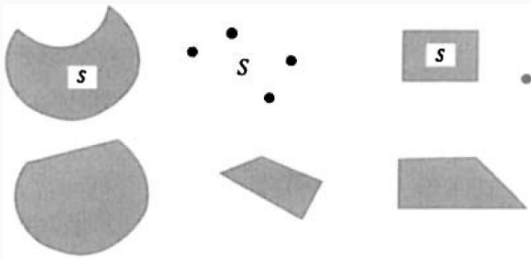
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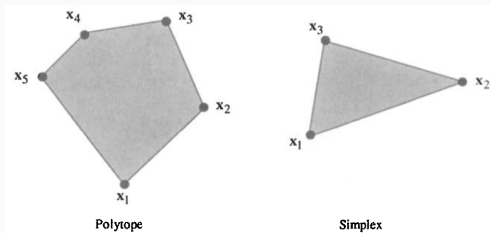
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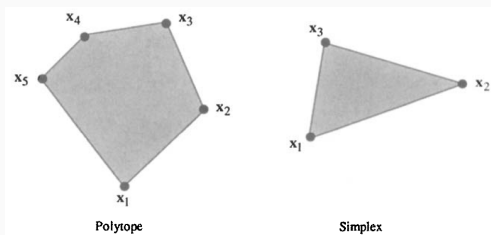
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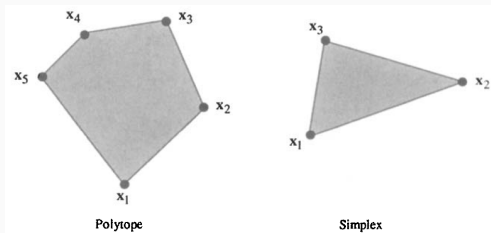


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Lemma

Let S be an arbitrary set in $\mathbb{R}^n$. Then, $\text{conv}(S)$ is the smallest convex set containing S . It is also the intersection of all convex sets containing S .

Carathéodory's Theorem (Convex Hull)

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Let S be an arbitrary set in $\mathbb{R}^n$. If $\mathbf{x} \in \text{conv}(S)$, then

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- Trivially holds for $\mathbf{x} \in S$.
- The choice of $\mathbf{x}_1, \dots, \mathbf{x}_{n+1}$ can depend on $\mathbf{x}$.

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- Is a ray a convex set?

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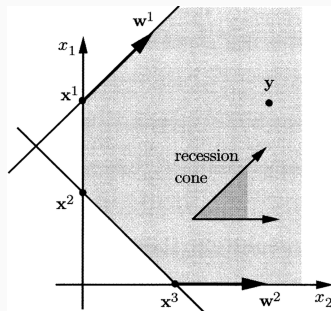
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Example: $X := \{\mathbf{x} \in \mathbb{R}_+^2 : x_1 - x_2 \geq -2, x_1 + x_2 \geq 1\}$



Summary and Next Steps

Key Takeaways

- Convex sets: line segment between any two points stays in the set
- Convex functions: function lies below the chord
- Operations preserving convexity: intersection, affine images, etc.
- Convex hulls and Carathéodory's theorem

Next Time:

- More on convex analysis: supporting hyperplanes, separation theorems
- Characterizations of convex functions

Questions?